# ORF 524 - Fall 2026

## Week 09

To motivate

$$\mathbb{E}[Y] = \underset{a \in \mathbb{R}}{\operatorname{argmin}} \; \mathbb{E}\left[(Y-a)^2\right] = \Psi_1(P_Y)$$

$$\operatorname{med}(Y) = \underset{a \in \mathbb{R}}{\operatorname{argmin}} \; \mathbb{E}\left[|Y-a|\right] = \Psi_2(P_Y)$$

$$\gamma_0 = \underset{\gamma \in \Gamma}{\operatorname{argmin}} \; -\mathbb{E}\left[\log f_Y(Y;\theta)\right]$$

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$$\theta_0 = \underset{\theta \in \Theta}{\operatorname{argmin}} \; Q(\theta) \quad ; \; Q(\theta) = \mathbb{E}\left[\rho(Y,\theta)\right]$$

$$\hat{\theta}_n = \underset{\theta \in \Theta}{\operatorname{argmin}} \; \hat{Q}_n(\theta) \quad ; \; \hat{Q}_n(\theta) = \hat{\mathbb{E}}_n\left[\rho(Y_i;\theta)\right] = \frac{1}{n}\sum_{i=1}^{n} \rho(Y_i;\theta)$$

$$\mathbb{E}[g(x)] = \int g(x)\, dF(x)$$

$$F(x) = \mathbb{E}[\mathbb{1}(x \le x)] = \mathbb{P}(x \le x)$$

$$\hat{\mathbb{E}}[g(x)] = \int g(x)\, d\hat{F}(x)$$

$$= \frac{1}{n} \sum_{i=1}^{n} g(x_i)$$

$$\hat{F}(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}(x_i \le x)$$

$$\theta_0 = \underset{\theta \in \Theta}{\operatorname{argmin}}\ Q(\theta) \quad ; \quad Q(\theta) = \mathbb{E}\left[\rho(Y, \theta)\right]$$

FoC:

$$0 = \frac{\partial}{\partial \theta} Q(\theta)\Big|_{\theta=\theta_0} = \mathbb{E}\left[\frac{\partial}{\partial \theta}\rho(Y, \theta)\right]\Big|_{\theta=\theta_0}$$

Plug-in analogue

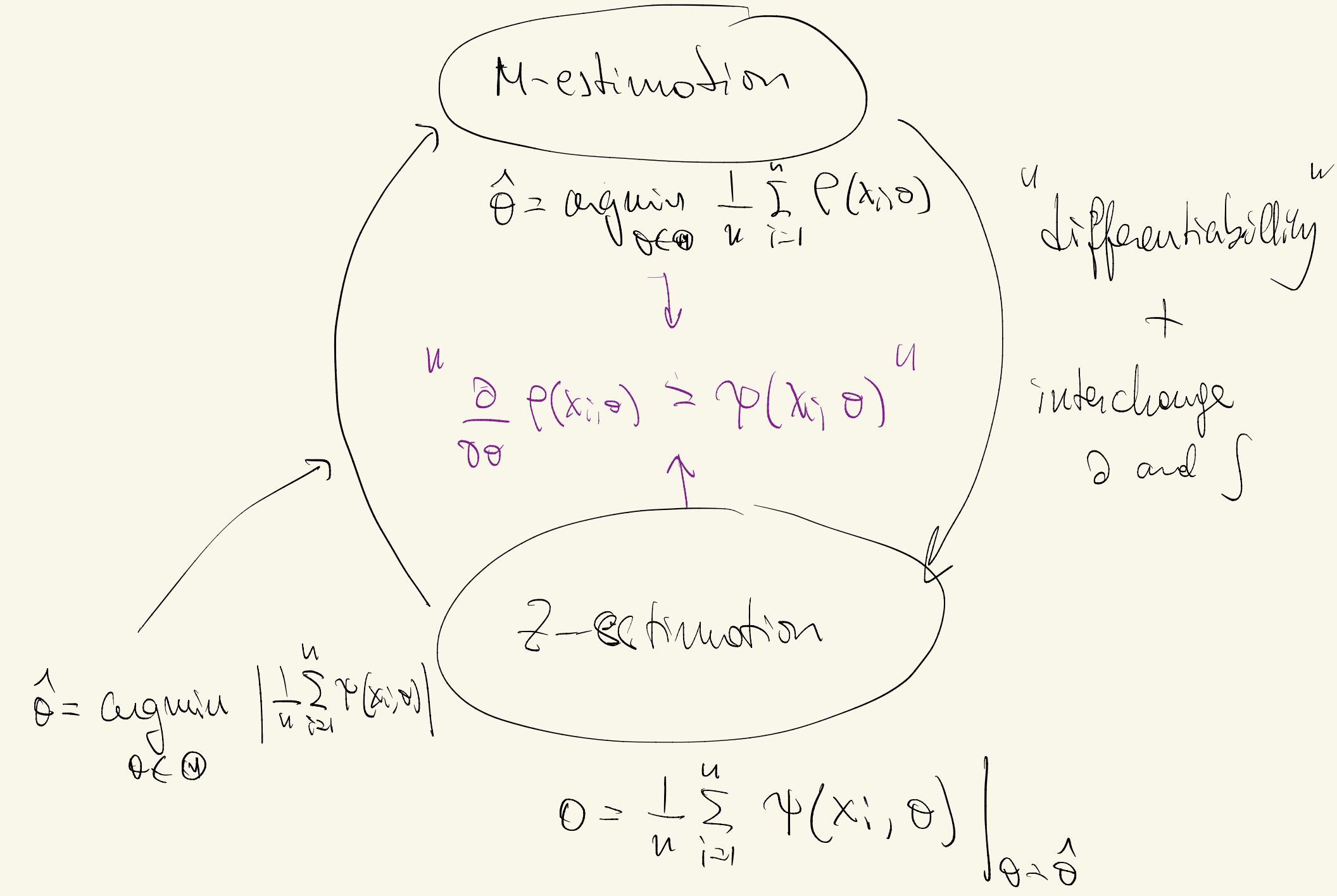
$$\hat{\theta}_n = \underset{\theta \in \Theta}{\operatorname{argmin}}\ \hat{Q}(\theta) \quad ; \quad \hat{Q}(\theta) = \hat{\mathbb{E}}_n\left[\rho(Y_i, \theta)\right]$$

M-estimation

FoC:

$$0 = \hat{\mathbb{E}}_n\left[\frac{\partial}{\partial \theta}\rho(Y_i, \theta)\right]\Big|_{\theta=\hat{\theta}} = \frac{1}{n}\sum_{i=1}^{n} \frac{\partial}{\partial \theta}\rho(Y_i, \theta)\Big|_{\theta=\hat{\theta}}$$

Z-estimation


M-estimation
$\hat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} \rho(x_i, \theta)$
"differentiability"
+
interchange $\partial$ and $\int$
$\frac{\partial}{\partial \theta} \rho(x_i, \theta) \doteq \psi(x_i, \theta)$
Z-estimation
$\hat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmin}} \left| \frac{1}{n} \sum_{i=1}^{n} \psi(x_i, \theta) \right|$
$0 = \frac{1}{n} \sum_{i=1}^{n} \psi(x_i, \theta) \Big|_{\theta = \hat{\theta}}$

# least squares

$$\hat{\beta} = \operatorname*{argmin}_{\beta \in B} \frac{1}{n} \sum_{i=1}^{n} \left( y_i - \Lambda(x_i'\beta) \right)^2$$

$$0 = 2 \cdot \frac{1}{n} \sum_{i=1}^{n} \left( y_i - \Lambda(x_i'\beta) \right) \cdot \Lambda^{(1)}(x_i'\beta) \cdot x_i \Big|_{\beta = \hat{\beta}}$$

$\Lambda(u) = u \Rightarrow$ OLS

$$\hat{\beta} = \left( \sum_{i=1}^{n} x_i x_i' \right)^{-1} \sum_{i=1}^{n} x_i y_i$$

$$\operatorname*{plim}_{n\to\infty} \hat{\theta} = \operatorname*{plim}_{n\to\infty} \operatorname*{argmin}_{\theta\in\Theta} \frac{1}{n}\sum_{i=1}^{n} \rho(x_i,\theta)$$

$$= \operatorname*{argmin}_{\theta\in\Theta} \operatorname*{plim}_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} \rho(x_i,\theta)$$

No!

$$= \operatorname*{argmin}_{\theta\in\Theta} \mathbb{E}\left[\rho(x_i,\theta)\right]$$

$$= \theta_0$$

Key (sufficient!): $\sup_{\theta\in\Theta} \left| \frac{1}{n}\sum_{i=1}^{n} \rho(x_i,\theta) - \mathbb{E}[\rho(x_i,\theta)] \right| = o_p(1)$

not necessary!

## Thm (Consistency – M-estimation):

$\hat{\theta}_n$ is an M-estimator. Two conditions hold:

(1) $$\inf_{\theta : \|\theta - \theta_0\| \geq \varepsilon} Q(\theta) > Q(\theta_0)$$

(2) $$\sup_{\theta \in \Theta} \left| Q_n(\theta) - Q(\theta) \right| \to_p 0$$

Then,

$$\|\hat{\theta} - \theta_0\| \to_p 0$$

Recall:

$$\hat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} \rho(x_i, \theta)$$

# Thm (Asy Linearity - Z-estimation)

(1) $\theta_0 \in \text{Int}(\Theta)$ ; $\hat{\theta} \to_p \theta_0$ ; $\Theta$ compact

(2) $\rho : \mathcal{X} \times \Theta \to \mathbb{R}$ is two times continuously differentiable in $\theta$.

(3) $\mathbb{E}\left[ \sup_{\|\theta - \theta_0\| \le \delta} |Q_n(\theta)| \right] < \infty$, similarly for derivatives.

(4) $\lambda_{\min}(H(\theta_0)) \ge c > 0$ ; $H(\theta_0) = H_0$

Then,

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi(x_i) + o_p(1) \rightsquigarrow N(0, V)$$

$$\mathbb{E}[\psi(x_i)] = 0 \quad , \quad 0 < \underbrace{\mathbb{E}[\psi(x_i)\psi(x_i)^T]}_{V} < \infty \quad ; \quad H_0 = \mathbb{E}\left[\frac{\partial}{\partial \theta \partial \theta^T} \rho(x_i; \theta_0)\right]$$

$$\psi(x_i) = -H_0^{-1} \frac{\partial}{\partial \theta} \rho(x_i, \theta_0) \implies V = H_0^{-1} \Sigma H_0^{-1}$$

"Proof"

$\hat{\theta} \to_p \theta_0$

$$0 = \frac{1}{n}\sum_{i=1}^{n} \frac{\partial}{\partial \theta} p(x_i, \theta)\Big|_{\theta=\hat{\theta}_n}$$

$$\tilde{\theta} = \lambda\hat{\theta} + (1-\lambda)\theta_0$$

$$= \frac{1}{n}\sum_{i=1}^{n}\left[\frac{\partial}{\partial \theta} p(x_i, \theta_0) + \frac{\partial}{\partial\theta\partial\theta^\tau} p(x_i, \theta)\Big|_{\tilde{\theta}}(\hat{\theta}-\theta_0)\right]$$

$$= \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{\partial}{\partial\theta} p(x_i, \theta_0) + \left[\frac{1}{n}\sum_{i=1}^{n}\frac{\partial}{\partial\theta\partial\theta^\tau} p(x_i, \tilde{\theta})\right]\sqrt{n}(\hat{\theta}-\theta_0)$$

then

$$\sqrt{n}(\hat{\theta}-\theta_0) = -\left[\frac{1}{n}\sum_{i=1}^{n}\frac{\partial}{\partial\theta\partial\theta^\tau} p(x_i, \tilde{\theta})\right]^{-1} \cdot \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{\partial}{\partial\theta} p(x_i, \theta_0)$$

Note: $\hat{\theta} \to_p \theta_0 \Rightarrow \tilde{\theta} \to_p \theta_0$, then

$$\frac{1}{n}\sum_{i=1}^{n} g(x_i, \tilde{\theta}) \to_p E[g(x_i, \theta_0)]$$

$$\left(g(x,\theta) = \frac{\partial}{\partial\theta\partial\theta^\tau} p(x;\theta)\right)$$

$\theta \mapsto \underbrace{\mathbb{E}[g(x_i,\theta)]}_{G(\theta)}$ is continuous.

$$\frac{1}{n}\sum_{i=1}^{n} g(x_i,\tilde{\theta})$$

$$= \left|\frac{1}{n}\sum_{i=1}^{m} g(x_i,\tilde{\theta}) - \mathbb{E}\left[g(x_i,\theta)\right]\Big|_{\tilde{\theta}}\right| + \mathbb{E}\left[g(x_i,\theta)\right]\Big|_{\tilde{\theta}} - \mathbb{E}\left[g(x_i,\theta_0)\right] + \underbrace{\mathbb{E}\left[g(x_i,\theta_0)\right]}_{\| \; G(\theta_0)}$$

$$\leq \sup_{\theta \in \Theta} \left|\frac{1}{n}\sum_{i=1}^{m} g(x_i,\theta) - \mathbb{E}[g(x_i,\theta)]\right|$$

by ULLN $\Rightarrow$ $o_p(1)$

$\Big\downarrow$ $o_p(1)$

by CMT.

$\tilde{\theta} \to_p \theta_0 \Rightarrow G(\tilde{\theta}) \to_p G(\theta_0)$

Note:

$$\mathbb{E}\left[g(x_i,\tilde{\theta})\right] = \int g(x_i(\omega);\tilde{\theta}(x_1(\omega),x_2(\omega),\dots x_n(\omega)))\, d\mathbb{P}^{\otimes}(\omega)$$

$$\mathbb{E}_{x_i}\left[g(x_i,\tilde{\theta})\right] = \int g(x_i(\omega),\tilde{\theta})\, d\mathbb{P}_{x_i}(\omega)$$

Then

$$\sqrt{n}\left(\hat{\theta}-\theta_0\right) \approx -\underbrace{\left[\mathbb{E}\left[\frac{\partial}{\partial\theta\partial\theta^T} p(x_i;\theta_0)\right]\right]^{-1}}_{H(\theta_0)} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \frac{\partial}{\partial\theta} p(x_i;\theta_0) + o_p(1)$$

$$= \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi(x_i,\theta_0) + o_p(1)$$

$$\psi(x,\theta) = -H^{-1}(\theta_0) \cdot \frac{\partial}{\partial\theta} p(x;\theta)$$

$$\mathbb{E}\left[\psi(x_i,\theta_0)\right] = -H^{-1}(\theta_0)\, \mathbb{E}\left[\frac{\partial}{\partial\theta} p(x_i;\theta_0)\right] = 0$$

$$\mathbb{V}\left[\psi(x_i;\theta_0)\right] = H^{-1}(\theta_0)\, \mathbb{V}\left[\frac{\partial}{\partial\theta} p(x_i;\theta_0)\right] \cdot H^{-1}(\theta_0)$$

Thus:

$$\sqrt{n}\left(\hat{\theta} - \theta_0\right) \rightsquigarrow N(0, V) \qquad V = H_0^{-1} \Sigma_0 H_0^{-1}$$

$$\Sigma_0 = \mathbb{E}\left[\frac{\partial}{\partial \theta} p(x_i, \theta_0) \frac{\partial}{\partial \theta^T} p(x_i, \theta_0)\right]$$

$$H_0 = \mathbb{E}\left[\frac{\partial}{\partial \theta \partial \theta^T} p(x_i, \theta_0)\right]$$

How do we estimate these people?

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^{n} \frac{\partial}{\partial \theta} p(x_i, \hat{\theta}) \frac{\partial}{\partial \theta^T} p(x_i, \hat{\theta})$$

$$\hat{H} = \frac{1}{n} \sum_{i=1}^{n} \frac{\partial}{\partial \theta \partial \theta^T} p(x_i, \hat{\theta})$$

$\left(g(x, \theta) = \frac{\partial}{\partial \theta} p(x, \theta) \frac{\partial}{\partial \theta^T} p(x, \theta)\right)$

$\Downarrow$ ULLN + CMT

$\hat{\Sigma} \to_p \Sigma$

Thus: an $100(1-\alpha)\%$ CI for $c^T\theta_0$ is

$$CI_{1-\alpha} = \left[ c^T\hat{\theta} - q_{1-\alpha/2} \cdot \sqrt{\frac{c^T \hat{H}^{-1} \hat{\Sigma} \hat{H}^{-1} c}{n}} \right.,$$

$$\left. c^T\hat{\theta} + q_{1-\alpha/2} \cdot \sqrt{\frac{c^T \hat{H}^{-1} \hat{\Sigma} \hat{H}^{-1} c}{n}} \right]$$

$$q_a = \Phi^{-1}(a) \quad ; \quad \Phi(a) = \int_{-\infty}^{a} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x^2\right) dx$$

$$\alpha = 0.05 \Rightarrow q_{1-\frac{\alpha}{2}} \approx 1.96$$

# Triviality of test

$H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$

- **Wald test** : $\sqrt{n}(\hat{\theta} - \theta_0) \rightsquigarrow N(0, V_0)$

$$T_n^W(\theta_0) \Rightarrow \sqrt{n}(\hat{\theta} - \theta_1) = \sqrt{n}(\hat{\theta} - \theta_0 + \theta_0 - \theta_1)$$

$$= \underbrace{\sqrt{n}(\hat{\theta} - \theta_0)}_{O_p(1)} + \underbrace{\sqrt{n}(\theta_1 - \theta_0)}_{\to \infty \text{ (absolute value)}}$$

$\Rightarrow |T_n^W(\theta_0)|$ rejects for large values.

$\Rightarrow$ Reject $H_0$ iff $|T_n^W(\theta_0)| \geq c$

Example : $\Theta \subseteq \mathbb{R}^q$

$$\|T_{n,p}^W(\theta)\|_q = \begin{cases} \|\sqrt{n}(\hat{\theta} - \theta_0)\|_p & p \in [1, \infty) \\ \|\sqrt{n}(\hat{\theta} - \theta_0)\|_\infty & p = \infty \end{cases}$$

# Rao / LM / Score test

Under $H_0 : \theta = \theta_0$,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \frac{\partial}{\partial \theta} \rho(x_i, \theta_0) \rightsquigarrow N(0, \Sigma_0)$$

If $\left| \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \frac{\partial}{\partial \theta} \rho(x_i, \theta_1) \right| \longrightarrow_p \infty \Rightarrow$ Non-trivial power -

then,

Reject $H_0$ iff $\left| T_n^{LM}(\theta_0) \right| > c$

# Deviance / LR test

$$Q_n(\hat{\theta}) - Q_n(\theta_0)$$

how different are they?

$$= \min_{\theta \in \Theta} Q_n(\theta) - Q_n(\theta_0)$$

this rejects for "small" (negative) values.